Quadratics: Expanding & Factorising

Practice test 7 · 40 marks · Higher tier · non-calculator

Name: _____ Date: _____ Score: _____ / 40

Section A — Expanding double brackets

A1. Expand and simplify (x + 7)(x – 2)

[2 marks]

A2. Expand and simplify (x – 1)(x – 12)

[2 marks]

A3. Expand and simplify (2x – 7)(3x – 1)

[2 marks]

A4. A square has sides of length (2x + 3) cm. Find an expression for its area. Give your answer in expanded form.

[2 marks]

Section B — Expanding three brackets

B1. Expand and simplify (x + 1)(x + 4)(x – 5)

[3 marks]

B2. Expand and simplify $(x + 3)(2x – 1)^2$

[3 marks]

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Factorise $x^2 + 14x + 45$

[2 marks]

C2. Factorise $x^2 – 2x – 63$

[2 marks]

C3. Factorise $49 – 4x^2$

[1 mark]

C4. $x^2 + bx – 20$ factorises to $(x – 4)(x + a)$. Find the values of a and b.

[2 marks]

Section D — Factorising $ax^2 + bx + c$

D1. Factorise $3x^2 + 14x + 8$

[2 marks]

D2. Factorise $5x^2 – 12x + 4$

[2 marks]

D3. Factorise fully $12x^2 + 22x - 4$

[2 marks]

Section E — Using factorising

E1. Solve $2x^2 - 7x + 3 = 0$

[3 marks]

E2. Simplify fully

$$\frac{x^2 - 3x - 10}{x^2 - 4}$$

[3 marks]

Section F — Problem solving & proof

F1. (a) Write $x^2 + 6x - 4$ in the form $(x + a)^2 + b$. (b) Hence write down the minimum value of $x^2 + 6x - 4$.

[3 marks]

F2. Prove that if you multiply two consecutive even numbers and add 1, the answer is always a square number.

[3 marks]

F3. Without a calculator, work out $76^2 - 24^2$

[1 mark]